Constructing Non-Koszul Quadratic Gorenstein Algebras I

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Disclaimer

We follow Section 4.2 of "Quadratic Gorenstein Rings and the Koszul Property I" by Matthew Mastroeni, Hal Schenck, and Mike Stillman (2020). **This is not my own work.**

Thanks to Adam LaClair and Mark Walker for helpful conversations.

Setup

Recall we are discussing graded algebras in this seminar of the form $R = S/I$ with $S = k[x_1, \ldots, x_n]$ and $I \subseteq S$ a homogeneous nondegenerate ideal. Let us assume that k is infinite [most of what we will discuss is invariant under flat base change anyway, so let's not quibble about semantics]. Most often R is CM, and let us also assume $\mathrm{char}(k) = 0$ (I will try to point out when this assumption is not needed).

Notation. The Nagata idealization of R with respect to a shifted canonical module is denoted

$$\widetilde{R} := R \ltimes \omega_R(-a-1)$$

where $a = \max\{j \mid H^d_{R_+}(R)_j \neq 0\}$ is the a-invariant of R. Recall we define the canonical module as a graded Hom module; $\omega_R =^* \mathrm{Hom}_k(H^d_{R_+}(R), k)$. Recall the **socle degree** is $\min\{i \mid R_i \neq 0\}$.

Write the statement of the next result on the side board before the talk.

Kesavan showed us the proof of

Theorem 1. *Let $R = S/I$ be a standard graded (Artinian) k-algebra.*

1. *If R is level, then $\widetilde{R}$ is a standard graded, (Artinian), Gorenstein k-algebra. In this case, one has* $\operatorname{codim}(\widetilde{R}) = \operatorname{codim}(R) + \operatorname{type}(R)$ *and* $\operatorname{reg}(\widetilde{R}) = \operatorname{reg}(R) + 1$. *(The equality comes from the fact that $\widetilde{R}$ is CM and considering the Hilbert series, artinian not required.)*

2. *If R is quadratic and superlevel, then $\widetilde{R}$ is quadratic. (Exploits (4), also without Artinian assumption).*

3. *If R is not Koszul, then $\widetilde{R}$ is not Koszul. (Needs only the Poincare series result of Gulliksen, no Artinian assumption).*

4. $\widetilde{R} \cong S[y_1, \ldots, y_t]/((I) + \mathcal{L} + (y_1, \ldots, y_t)^2)$, *where* $t = \operatorname{type}(R)$ *and*

$$\mathcal{L} = \left(\sum_{i=1}^{t} f_i y_i \mid (f_1, \ldots, f_t) \in \operatorname{Syz}_1^S(\omega_R) \right).$$

(This is all valid without the artinian assumption.)

Reading data from h-vectors

The **h-vector** of R is $h(R) = (h_0, h_1, \ldots, h_r)$, where

$$H_R(t) = h_R(t)/(1-t)^d$$

is the Hilbert series of R and $h_R(t) = \sum_{i=0}^{r} h_i t^i$. Recall that when R is Cohen-Macaulay, we can read the regularity of and codimension off of the h-vector:

$$r = \operatorname{reg}(R) \text{ and } \operatorname{codim}(R) = h_1.$$

Moreover, when R is superlevel (i.e., ω_R is generated in a single degree and has a linear presentation over R) one also obtains $h_r = \text{type}(R)$.

Our Main Question

Question 1. *Fix positive integers c and r. Is every quadratic Gorenstein ring R with codimension c and regularity r Koszul?*

This question is still open for $r = 3$ and $c = 6, 7$.

Ideals of generic forms

This section is where we need $\text{char}(k) = 0$.

Definition 1. *By a generic set of g quadrics, we mean a point in the Zariski open subset of S_2^g.*

To give a family of examples, we will take the following results of Fröberg and Löfwall and Hochster and Laksov without proof.

Fact 1 (Fröberg-Löfwall 1997)**.** *If $R = S/I$ is an Artinian algebra with I generated by g generic quadrics in n variables, then R is Koszul iff $g = n$ or $g \geq \frac{n^2+2n}{4}$.*

Remark. *Their result is more general than this. The proof relies on some computations involving Grassmanians, the Bar complex, and Koszul duality.*

Fact 2 (Hochster-Laksov (1987))**.** *For an ideal I generated by g generic forms of degree d in n many variables, one has*

$$\dim_k I_{d+1} = \min\left\{gn, \binom{n+d}{d+1}\right\}.$$

Remark. *I couldn't get access or find this paper, so I have nothing to share regarding this proof.*

Thus, when $R = S/I$ is defined by g generic quadrics (in n variables) is Artinian, non-Koszul, and has socle degree 2 (this should imply $d = 1$ above b/c then $d+1 = 2$ is quadratic) if and only if

$$\frac{n^2+3n+2}{6} \leq g < \frac{n^2+2n}{4} \qquad (\star)$$

(notice the quadratic in the lower bound is so that when multiplying the left two terms by n we achieve the minimum in Fact 2). The h-vector of S in this case is $(1, n, \binom{n+2}{2} - g)$, because R has socle degree two, $H_R(t) = \dim_k R_0 + \dim_k R_1 t + \dim_k R_2 t^2$ and $R_0 = k$, R_1 is dimension n as a k-vs, and $\binom{n+2}{2} - g$ is the number of degree two monomials minus the g many that we killed.

With these bounds, the authors propose the following.

Theorem 2. *Let $I \subseteq S = k[x_1, \ldots, x_n]$ be an ideal generated by g generic quadrics, with $n \geq 4$ and g satisfies the inequalities in $(\star)$. Then $R = S/I$ is non-Koszul and superlevel. Consequently, $\widetilde{R}$ is a non-Koszul quadratic Gorenstein ring with h-vector*

$$h(\widetilde{R}) = (1, \frac{n^2+3n}{2} - g, \frac{n^2+3n}{2} - g, 1). \qquad (\dagger)$$

Notice that the consequence (except maybe the h-vector computation) comes almost for free from Theorem 1.

Before we see the proof, one may wonder if all non-Koszul socle degree 2 regularity 3 algebras come from this idealization construction. It is an example of Roos that shows this is not the case.

Example 1 (Roos 1993)**.** *$S = k[x, y, z, w, u, v]$ (and $\operatorname{char}(k) = 0$) with*

$$I = (x^2, y^2, z^2, w^2, u^2, v^2, xy, yz, uv, vw, xz + 3zw - uw, zw + xu + uw).$$

Then $R = S/I$ is superlevel and socle degree two (exercise) and not Koszul, as can be seen

in the Betti table (likely computed by Macaulay 2).

	0	*1*	*2*	*3*	*4*	*5*	*6*
0	*1*	-	-	-	-	-	-
1	-	*12*	*16*	*2*	-	-	-
2	-	-	*32*	*96*	*100*	*48*	*9*

The h-vector of this algebra is $(1, 6, 9)$. *Notice* R *is Artinian since any prime that contains* I *contains* $\mathfrak{m}$. *The first two numbers are easy to see since* S *is a* 6 *variable polynomial ring over* k, *and* 9 *can be computed by counting the number of two forms and subtracting 12 for the number of forms we killed.*

Another application of Theorem 1 shows that $\widetilde{R}$ *is a quadratic Gorenstein non-Koszul algebra with h-vector*

$$(1, \underbrace{15}_{=\operatorname{codim}(R)+\operatorname{type}(R)=6+9}, 15, 1)$$

since Gorenstein implies a symmetric h-polynomial and superlevel, $\operatorname{type}(\widetilde{R}) = 1$, *and* $\operatorname{reg}(\widetilde{R}) = 1 + \operatorname{reg}(R)$. *Thus,* (†) *does not characterize all h-vectors of quadratic Gorenstein rings of regularity* 3, *because no choice of* n *and* g *can yield* 15 *in* (†).

So one may ask what h-vectors are possible for non-Koszul quadratic Gorenstein k-algebras of socle degree 2 and regularity 3? Here's a (partial) answer.

Theorem 3. *(Over a field of characteristic zero) There exist non-Koszul quadratic Gorenstein rings with h-vector* $(1, c, c, 1)$ *for all* $c \geq 9$. *(Notice this answers our main question in the negative for* $r = 3$ *and* $c \geq 9$.*)*

Recall, the h-vector must have length 3/4 because of regularity 3, must be symmetric (so the first and last numbers are one) and the second middle numbers are the codimension.

Proof Sketch. For each fixed n, it is clear that the codimension function (defined from Thrm

2)

$$c(n,g) = \frac{n^2+3n}{2} - g$$

is decreasing in g and takes every integer value between

$$c(n, \underbrace{g_{\max}(n)}_{\lceil \frac{n^2+2n}{4} \rceil - 1}) \leq c \leq c(n, \underbrace{g_{\min}(n)}_{\lceil \frac{n^2+3n+2}{2} \rceil}).$$

Hence, there shall be no integer gaps in the codimensions so long as

$$c(n, g_{\min}(n)) + 1 \geq c(n+1, g_{\max}(n+1)).$$

This holds for all $n \geq 9$ via the estimate

$$\begin{aligned} c(n, g_{\min}(n)) - c(n+1, g_{\max}(n+1)) &> c(n, \frac{n^2+3n+2}{6} + 1) - c(n+1, \frac{(n+1)^2+2(n+1)}{4} - 1) \\ &= \frac{n^2-6n-43}{12} \\ &\geq -2, \end{aligned}$$

and for $n = 8$ via a direct computation.

Thus, the construction in Thrm 2 gives non-Koszul quadratic Gorenstein rings with h-vector $(1, c, c, 1)$ for all $c \geq 9$ except for $c \in \{10, 11, 14, 15, 18, 19, 24\}$. Example 1 gives $c = 15$ case, and the authors claim you can make slight modifications to get superlevel non-Koszul algebras of socle degree two in the remaining cases. They arise from ideals of the form $I = J + L$, where J is a rather tedius ideal to write down in each case and L is generated by the squares of variables appearing in the generators of J. Those quotients S/I are then easily checked in M2 to be superlevel with idealization having the appropriate h-vector $(1, c, c, 1)$ for each c in the above list. $\square$

Tensor products

To provide further examples, we need some results about tensor products. These are motivated by the following observation.

Remark 1. *When R is a non-Koszul quadratic Gorenstein ring, then the idealization*

$$\begin{aligned} \widetilde{R} &\cong S[y]/((I)+(y^2)) \qquad \text{(Theorem 1(4))} \\ &\cong R \otimes_k k[y]/(y^2). \end{aligned}$$

will again be non-Koszul, quadratic, and Gorenstein with codimension and regularity increased by one [this follows from Theorem 1(4), no Artinian assumption needed].

Notice that $A = k[y]/(y^2)$ is Koszul (Dorian gave this example in CARS in fact, the minimal free res is $\cdots \to A(-2) \xrightarrow{y} A(-1) \xrightarrow{y} A \to 0$) of codimension one and type one (it's ci hence Gorenstein).

Proof. This follows almost immediately by Theorem 1. Notice every Gorenstein ring is superlevel since $\omega_R \cong R(a)$. Now if you recall the proof Keseavan gave us, removing Artinian from the assumption in Theorem 1, one can still obtain (1) without the Artinian conclusion.

Further, the proof of (2) did not rely on the Artinian assumption, it merely used the isomorphism in (4) which held in general as well. Finally, (3) held without the Artinian assumption as well [it followed from a result of Gulliksen by messing around with Poincare series]. □

We now can get more results for the negative answer to our question.

Proposition 1. *Over a field of characteristic zero, there exists a non-Koszul quadratic Gorenstein ring of codimension c and regularity r for every $r \geq 6$ and $c \geq r+3$.*

Lemma 1. *Let $R = S/I$ be a quadratic Gorenstein ring and B a superlevel Koszul alegbra. Then $R' = R \otimes_k B$ is Koszul (resp. level, superlevel) if and only if R is. Further, we have*

$$\begin{aligned}\operatorname{codim} R' &= \operatorname{codim} R + \operatorname{codim} B \\ \operatorname{type} R' &= \operatorname{type} R \cdot \operatorname{type} B \\ \operatorname{reg} R' &= \operatorname{reg} R + \operatorname{reg} B.\end{aligned}$$

We ran out of time to prove these.

Next time...

- Proof of Theorem 2
- Proof of Prop 1
- Proof of Lemma 1
- More examples (almost complete intersections)
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